



Milton Education Village

District Energy Feasibility Study – Proposal

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About GEOptimize

The use of fossil fuels to heat and cool buildings is being restricted around the world. Building and vehicle electrification is forcing more efficient use of electricity. Ground source heat pump (GSHP) systems are powered with electricity to deliver heating and cooling needs more efficiently than any other available technology.

GEOptimize focuses on optimizing the cost and performance of GSHP systems in homes, commercial, and industrial buildings. GEOptimize also focuses on large scale district geothermal grids that take advantage of waste energy from refrigeration systems, wastewater, combined heat and power, and renewable energy resources.

Conventional heating and cooling systems are connected to a gas or electric utility. Designers are typically not concerned about the amount of energy needed during a year, but only with the instantaneous peak heating and cooling loads.

A GSHP system is connected to a ground heat exchanger that takes advantage of the energy storage capacity of the earth. To determine how it will perform, a designer needs to know:

- The amount of energy the GHX must deal with instantaneously and annually.
- Thermal properties of the earth the GHX is built in.
- The land area available to construct the GHX.

GEOptimize uses commercially available software to model building performance and works closely with the design team to optimize the building and mechanical systems to perform well with a GHX for the life of the building. For larger systems with multiple buildings and other energy resources, GEOptimize has developed customized modules using TRNSYS to simulate the performance of district systems such as university campuses and community energy sharing systems. GEOptimize is certified to provide professional engineering services in the province of Ontario.

GEOptimize has developed an extensive library of hourly building energy models of many building types. These are modified to simulate the performance of your proposed building project with a GSHP system and compare it to other types of systems from an economic, technical, and environmental perspective to quickly assess their feasibility.

When constructed, GEOptimize works with you and your building operator to develop an understanding of how the system is designed to perform and to continue optimizing system performance into the future. Ongoing monitoring of your system is essential to allow for changes in the building, how it is used and operated and compensate for changes in the weather from year to year. GEOptimize strongly recommends that all GSHP systems are continuously monitored to ensure optimum performance.

Milton Education Village Project Background

Mattamy Homes is planning to develop a 640-acre parcel of land in Milton, ON, Canada called the Milton Education Village (MEV) (Figure 1). The parcel of land is split into several ownership groups including Mattamy, Laurier University, Conestoga College, Milton Land Four, Schlegel Village, Hospice, and the Town of Milton. The Mattamy lands is planned for a 10 to 15 year build out and the education campuses are on a proposed 30+ year build out schedule. The buildings include a wide range of space

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usage types including office, multi-family, retail, and education. Sustainability is a new core value to Mattamy and the MEV is a unique, innovation focused, showcase development.



Figure 1: MEV Preliminary Master Plan

District Energy Systems Overview

Distribution Configurations

District systems have evolved over time since the earliest systems were built in the late 1800's (Figure 2). The most significant change has been a reduction in temperatures of the heat transfer fluid delivered to buildings connected to it. As delivery temperatures have dropped the systems can take advantage of low-grade waste heat sources (heat recovery from building cooling & refrigeration systems, wastewater, etc.) and renewable energy sources has increased overall system efficiency. Figure 2 illustrates the steady increase in system efficiency as the temperature of the heat transfer fluid to connected buildings has decreased since the late 1800's.

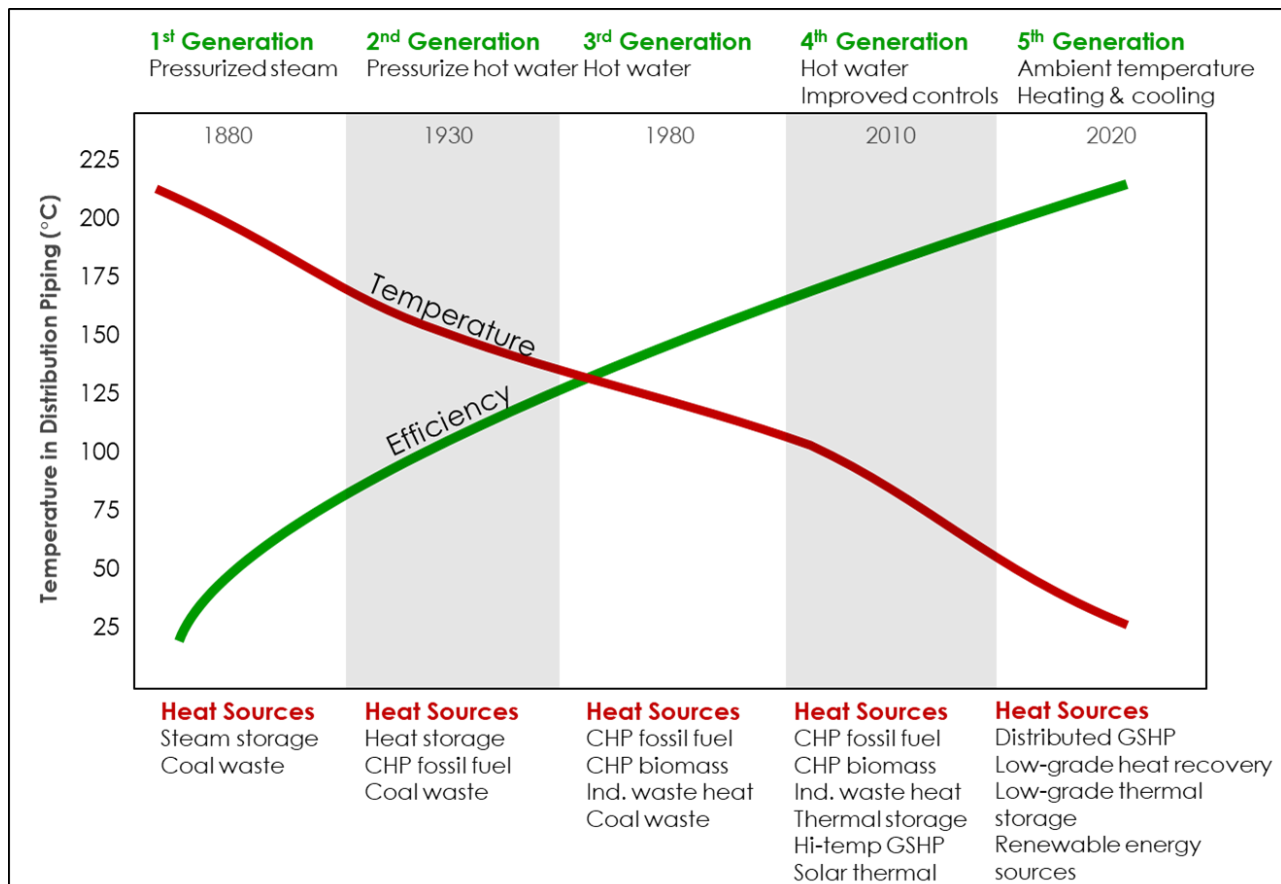


Figure 2: History of district energy systems and temperature vs. efficiency

4th generation district energy systems (DES's) are designed with a central energy plant that produces hot and chilled water. Two pairs of insulated supply and return pipes circulate hot and chilled water from a central energy plant to each building on the DES. If a building requires heat, hot water is pumped from the hot water supply line to the building distribution system. The temperature of the hot water is cooled as the building is heated and is pumped back to the central energy plant from the return line. Similarly, when a building is cooled, heat is rejected from the building to the chilled water and returned to the central energy plant through the return pipe (Figure 3).

Milton Education Village – District Feasibility Study

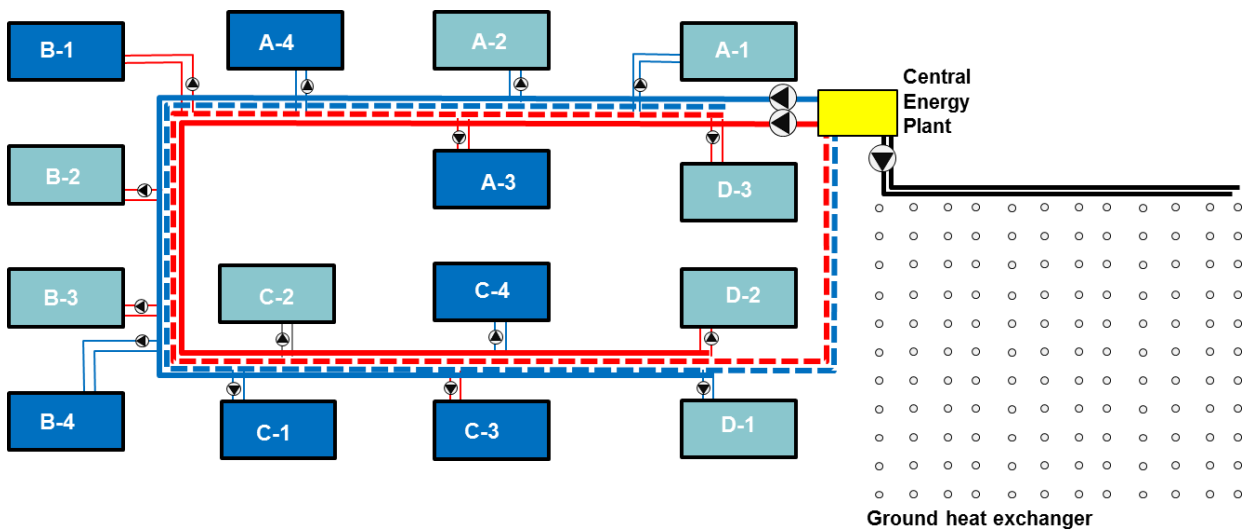


Figure 3: 4G district GSHP energy system

The main disadvantages of a 4th generation District Energy Systems (DES's) are:

- Distribution piping and circulation pumps must have the capacity to meet the combined peak heating and cooling loads of all buildings connected to the DES.
- DES delivers same water temperature to all buildings. The building with the highest water temperature governs the supply water temperature of the entire system reducing overall system efficiency.
- Requires full planning of future system to ensure central plant has capacity to meet heating and cooling demands of completed development although development construction may not be completed for several years.

5G DES's takes advantage of energy sharing by moving energy from a building that is cooling and rejecting heat to the system to another building that is extracting heat from the system. When the amount of heat rejected to the energy transfer pipe (ETP) is equal to the amount of heat extracted from the ETP, energy does not need to be added to or removed from the system. When the temperature of the ETP falls outside normal operating parameters (4°C/40°F to 29°C/85°F) the energy source is activated to maintain the normal operating temperatures. Rather than using a boiler as the energy source or cooling tower to dissipate heat, a pump connected to a ground heat exchanger (GHX) is activated to add or remove energy from the ETP (Figure 4).

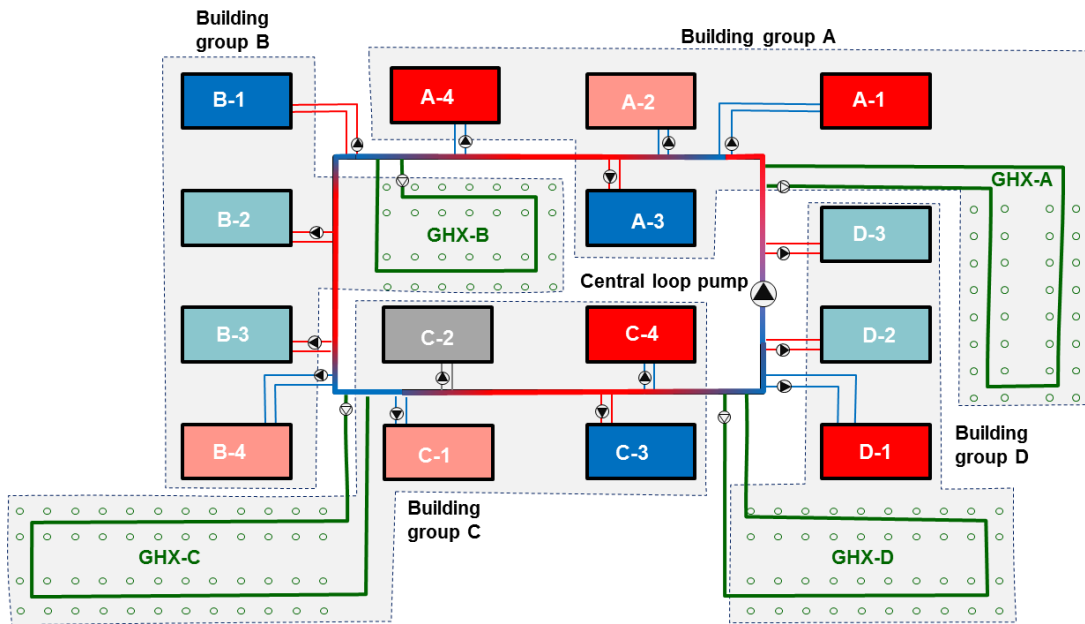


Figure 4: 5G district GSHP energy system

The main advantages of 5G DES's are:

- A single ETP operates at GHX temperature and does not require insulation as delivery temperature to buildings is not critical, reducing initial capital cost.
- Mechanical systems in each building are designed and optimized for the distribution system in the building without compromising the efficiency of the entire system. Buildings can be connected to the ETP at any location without affecting system operation which allows the system to adapt to changes as the needs of the client change.

Heat Sources & Sinks

Vertical & Horizontal Closed Loop Ground Heat Exchangers

Vertical GHX's consist of a plastic HDPE U-bend that can be installed into the ground to a depth up to 1,000 ft. The benefits to a vertical GHX are if you have restricted land area, you can increase your heat exchange surface area by drilling deeper vertically rather than using land horizontally.

Horizontal directionally drilled (HDD) GHX's consist of drilling a pilot hole with a directional drilling rig and then pulling a U-bend back through the pilot hole (Figure 5). The drill rig can install piping as deep as 30 ft and can have 3 sets of u-tubes stacked vertically to a depth of 30 ft, 20 ft, and 10 ft, for example. There are several advantages of an HDD GHX. These include:

- Lower capital costs
- Minimal site disturbance (horizontal boreholes can be installed under existing paved parking areas or even under buildings)
- The proximity of the horizontal GHX piping to the atmosphere allows dissipation of excess heat to the atmosphere in winter and warm outdoor air temperatures and rainfall can add energy to the GHX during the summer, reducing the potential for long-term temperature degradation

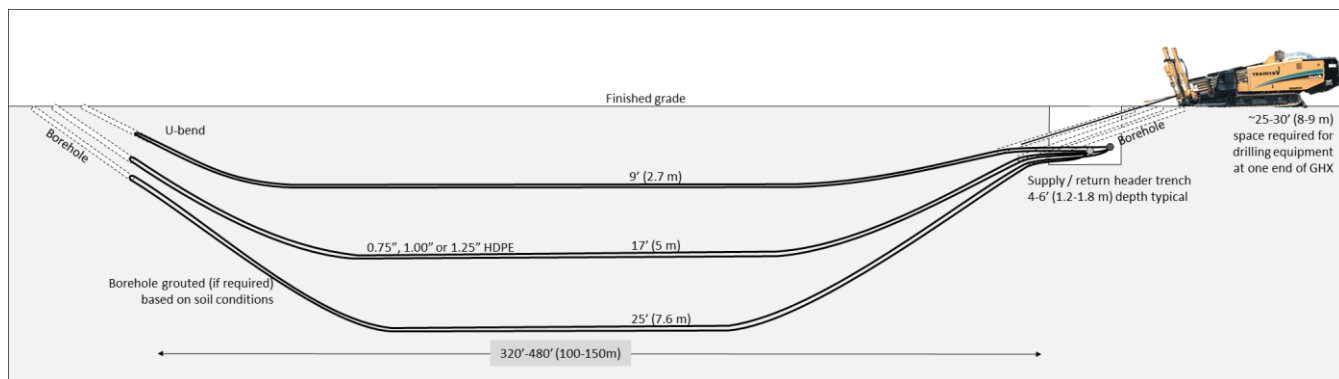


Figure 5: Visual representation of horizontal directionally drilled boreholes.

Open Loop Ground Heat Exchanger

Groundwater can provide a constant heat source / sink when it is available. The availability of ground water can be determined through a pump test. Groundwater in the Milton area is at a constant temperature of approximately 50°F.

Groundwater can be drawn from a supply well, circulated through a heat exchanger connected to the district energy system (DES) and returned to the ground via a return well. Alternatively, a heat exchanger can be installed directly in a water well. Inducing flow across the heat exchanger by drawing water from a water bearing zone at one depth and returning it into a water bearing zone at a different level above a heat exchanger installed inside the water well. Utilizing an open loop GHX can help increase the overall efficiency of the DES by moderating loop temperatures throughout the year.

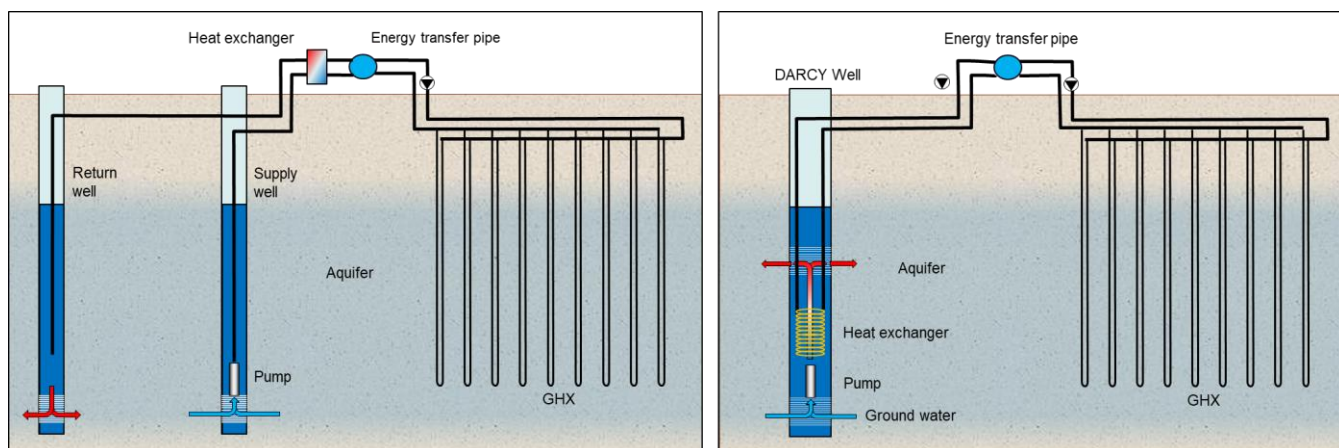


Figure 6: Supply / return well pair with out of heat exchanger and supply/return well pair with the installation of a heat exchanger located in the well casing.

Surface Water Heat Exchanger

Surface water heat exchanger (SWHX) systems could be an alternative or additive technology to support the DES. They utilize surface water bodies such as storm water ponds as heat sources / sinks. Closed loop SWHX use submerged heat exchangers to extract or reject heat from the surface water body. Heat transfer between the heat exchanger and the lake is typically due to natural convection. For a SWHX, the heat exchanger coils are often made of high-density polyethylene (HDPE) pipes or

titanium or stainless-steel heat exchangers. There are two main configurations of surface water heat exchangers:

- HDPE slinky type (Figure 7)
- Stainless steel plate (Figure 8)



Figure 7: HDPE slinky surface water heat exchanger being sunk to the bottom of a pond.



Figure 8: Stainless steel plate surface water heat exchanger.

Wastewater Energy Transfer

The use of wastewater as a heat source/sink has been overlooked as an energy saving measure in modern day mechanical designs. Recent technologies have been developed for extracting and injecting heat into a wastewater line. Two new technologies include the *Sharc* wastewater system and the *@Source-Energy* pipe. The *@Source-Energy* pipe is a hollow concrete cylinder with a helical high-density polyethylene pipe embedded within the concrete pipe walls that has an outlet and inlet on each length of pipe, which are attached to a heat pump. Heat is captured from effluent in sanitary and storm sewer pipes, and from the adjacent ground (Figure 9). Since wastewater is constant at approximately 60 to 70°F (15 to 20°C) year-round, energy for the DES can either be extracted or

rejected to the wastewater. Another option is the *Sharc* system which can be installed where wastewater exits to the wastewater treatment plant and transfers heat from the wastewater to a holding tank connected to the district energy system. The benefit to this system is that it can be tied into the existing sewage pipe system.



Figure 9: Wastewater heat recovery system integrated with vertical GHX.

Heat Rejection

Cooling towers are expensive and relatively useless other than their function of dissipating heat. Irrigation lines for football fields, soccer pitches, and baseball diamonds can be integrated into the DES with a water-to-water heat exchanger to get rid of excess heat (Figure 10). This allows for further control of the loop temperature and a useful dissipation of heat. For the remainder of the year, snow melt systems can be utilized to act as a heat dissipation device that also reduces the need for paid snow removal. Piping is laid under sidewalks or parking lots and connected to the DES (Figure 11). The piping can be installed when renovations are needed on the sidewalk or parking lots after years of deterioration.



Figure 10: Irrigation lines for football fields, soccer pitches, and baseball diamonds.

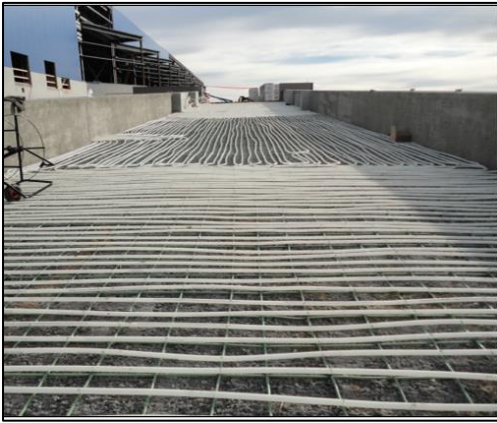


Figure 11: Piping under sidewalks and parking lots for snow melt.

Combined Heat and Power

Combined heat and power plants (CHP) burn fossil fuels or biomass to produce electricity. If 100 units of energy are consumed, the CHP produces approximately 30 units of electricity, 45 units of waste heat that is easily recovered, and 25 units of heat that is released to the atmosphere of exhaust gases, for an overall combined efficiency of approximately 75%. The waste heat of a CHP can potentially be connected to the DES and transferred to the buildings and / or GHX modules.

Power Generation

Solar Photovoltaic

Photovoltaic (PV) materials and devices convert sunlight into electrical energy. A single PV device is known as a cell. An individual PV cell is usually small, typically producing about 1 or 2 watts of power. To boost the power output of PV cells, they are connected together in chains to form larger units known as modules or panels. Modules can be used individually, or several can be connected to form arrays. One or more arrays is then connected to the electrical grid as part of a complete PV system. PV modules and arrays are just one part of a PV system. Systems also include mounting structures that follow the sun, along with the components that take the direct-current (DC) electricity produced by modules and convert it to the alternating-current (AC) electricity used to power all the appliances in your building.

Wind Power

A wind turbine generates electricity using wind to spin a generator. The rotor connects to the generator, either directly or through a shaft and a series of gears that speed up the rotation and allow for a physically smaller generator. This translation of aerodynamic force to rotation of a generator creates electricity.

Battery Energy Storage System

Wind and solar energy are intermittent and inconsistent. Due to this inconsistency, a battery energy storage system (BESS) eliminates intermittency using batteries to store and distribute energy as it is needed (Figure 12). At its most basic level, a BESS consists of one or more batteries that store electrical energy for use when needed. This stored energy can then be drawn upon when needed to meet various demands for power across different applications. BESS's can also provide advantages over other energy storage systems, including greater efficiency and flexibility, faster response times when powering equipment or devices, and lower costs overall. Lithium-ion batteries are one of the

most popular forms of energy storage in the world, accounting for 85.6% of deployed energy storage systems in 2015. Li-ion batteries consist of lithium metal oxides in the positive electrode, where lithium ions can be stored, and carbon in the negative electrode. During charging, lithium ions pass from the positive to the negative electrode, while the opposite occurs during discharging.



Figure 12: Battery energy storage system with solar PV array and wind power.

Integrating renewable energy production and storage with the energy efficiency of a GSHP system reduces the capacity and cost of both the energy production technology required and the energy storage system needed to support it.

Proposed Study Methodology

Building Energy Modelling

GEOptimize has developed an extensive library of hourly building energy models of many building types. These can be matched to the individual space usage and scaled to the building area to determine the individual building loads or the aggregated DES loads. Iterations of the energy model are run to determine the impact of a range of energy efficiency measures on the annual energy load balance and the proposed GHX design. Some of the iterations *may* include:

- Mattamy's current standard building performance vs. updated building performance
- Calculation and inclusion of domestic hot water to reduce heat rejection to the GHX
- Exhaust air energy recovery system to reduce overall heating and cooling loads
- Unique heating and / or cooling loads that may become apparent during design development

The information will be used to create a set of iterative hourly energy models that may be used to inform the design with recommendations of changes that can potentially be made to the building and / or mechanical systems that will result in an increase of energy efficiency. The building energy models will be used to approximate the peak and annual electrical energy consumption to size a clean power generation system within a rough order of magnitude. The building energy models will also be used to determine the energy costs with the projected carbon emissions for 3 district energy system options. The options will include: a 'baseline' district energy system which would be the least energy efficient, a mid-efficiency district energy system, and a high efficiency district energy system. The technology for the options will be decided in collaboration with Mattamy with the high efficiency district energy option being a GSHP system.

District Energy Modelling

The building energy models developed in the initial phase will be used to determine the heating and cooling demands of the various plots of land. The energy usage for each option will be determined based on ASHRAE standard efficiencies for the heating and cooling technology decided. Capital and operating costs will be based on unit pricing from local utilities and contractors in terms of \$/kWh or \$/therm for the operating costs and \$/ton for capital costs including supply and installation of the mechanical equipment. The capital costs associated with the GHX will be estimated based on interviews with local drilling contractors at a rate of \$/ft for supply and installation. GEOptimize will work with Mattamy throughout the pricing process. The carbon emissions will be based on the energy usage and the carbon intensity of the electricity generation options on site. From this analysis, a high level LCOE comparison will be developed. Once the 3 DES's are determined, site plans will be developed and GEOptimize will work with the project team to develop an implementation plan based on a modular, scalable, and flexible approach.

High Level District Geothermal System Modelling

For the high-efficient district energy system option, high level district geothermal system modelling will be performed. The type and configuration of GHX that will be selected for the DES will be determined based on a review of the local geology, hydrogeology, project site and drilling space available. Soil thermal properties will be determined by reviewing available drilling logs and interviews with geologists and / or drilling contractors familiar with the region. A series of simulations of the model are developed to optimize system performance including:

- The size and configuration of GHX modules.
- Location of GHX modules in relation to building heating and cooling requirements.
- Additional waste and renewable energy resources and heat sinks and their impact on the size and performance of the GHX modules.

The ideal combination of heat sources and heat sinks will be identified based on this information. When the optimal type or combination of types of GHX have been identified, the developed energy loads will be imported into Ground Loop Design (GLD2016) software to model the preliminary size of the GHX for each district module based on the aggregated district hourly load set. The preliminary size of the GHX will be split into areas where drilling is feasible and identified on a campus map.

Collaboration Workshops

An important step in this process is to work with Mattamy to create a path forward for the district heating and cooling system implementation for the MEV development. Throughout the analysis, 4 in-person workshops will be organized at specific milestones throughout the project. The workshops will occur at the following milestones:

- Beginning of the project for data collection and methodology discussion
- Building energy modelling completion
- District energy modelling completion
- Final report completion

Deliverables

Throughout the analysis period, there will be 4 in-person workshops at the Mattamy headquarters to discuss the project and have knowledge sharing sessions about energy efficiency, ground source heat pump systems, and discuss the progress of the analysis.

A report detailing the following:

- Aggregated energy load sets split into land ownership groups based on representative models, square footage, and space usage.
 - Iterations of the energy models will be developed based on efficiency measures including Mattamy's current standard building performance vs. proposed building performance.
- Analysis of clean power generation options for the development.
- Analysis of a 'baseline', mid-efficiency, and high efficiency district energy system including advantages vs. disadvantages.
- Site plans for a 'baseline', mid-efficiency, and high efficiency district energy system including required land area for GHX's in relation to the civil infrastructure including roads and buildings.
- Develop an implantation plan based on gathered information and building construction priority.
- Budget capital and operating cost estimates for a 'baseline', mid-efficiency, and high efficiency district energy system
- LCOE of a 'baseline', mid-efficiency, and high efficiency district energy system
- On and off-site annual carbon forecast for a 'baseline', mid-efficiency, and high efficiency district energy system
- Final recommendations for applying the system at a large scale.

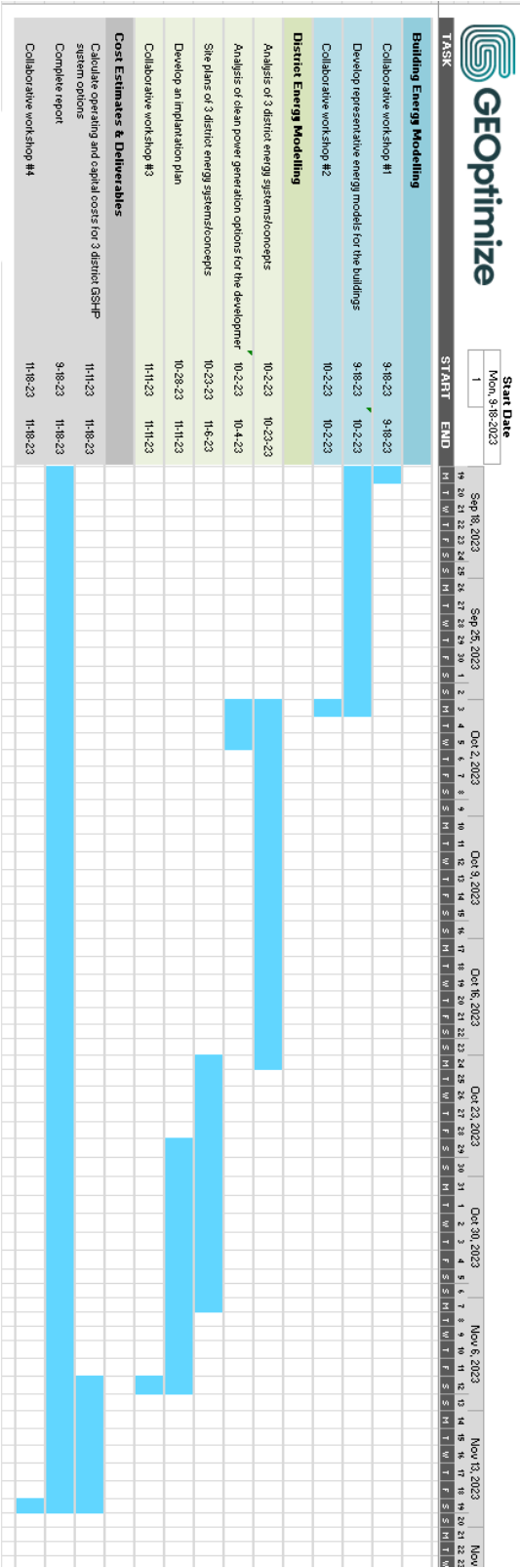
The report will also be summarized in a PowerPoint presentation that will be presented to the project team in-person at the Mattamy headquarters.

Pricing

Description	Rate (\$/hr)	Hours	Line Total
Representative hourly energy model for building at the proposed site	\$250.00	32	\$8,000.00
Analysis of a baseline, mid-efficiency and high efficiency district energy systems/concepts	\$250.00	40	\$10,000.00
Analysis of clean power generation options for the development	\$250.00	8	\$2,000.00
Site plans of baseline, mid-efficiency and high efficiency district energy systems/concepts	\$250.00	16	\$4,000.00
Develop an implantation plan	\$250.00	32	\$8,000.00
Costing and CO ₂ estimates	\$250.00	12	\$3,000.00
Final analysis and presentation of written report	\$250.00	32	\$8,000.00
Collaborative in-person workshops	\$2,000.00	4	\$8,000.00
Subtotal (CAD)			\$51,000.00
	+ON HST	13%	\$6,630.00
Total (CAD)			\$57,630.00

Work Plan

Timeline is dependent on receiving all required information at the beginning of the project. Delays will be prevalent if data collection is delayed.



Key Staff



Ed Lohrenz, BES, CGD

Ed has worked in the ground source heat pump industry since the early 1980's. He has been fortunate to have worked with many people as passionate about the technology as he is. He has had the opportunity to design GSHP systems for many applications and in climates ranging from the Australian outback to the Siberian tundra as well as many unique projects across Canada and the U.S.



Stephen Hak, EIT, BSc

Stephen is a Biosystems Engineering graduate from the University of Manitoba. He is currently pursuing a Master of Science in Engineering with a focus in geothermal energy. He has gained significant experience using TRNSYS for simulating the performance of ambient temperature, multi-building district geothermal systems, and experience in energy modelling, ground heat exchanger design, and ground source heat pump systems.



Ian Lohrenz, P.Eng, CGD

Ian has focused on energy efficiency in buildings since graduating from the University of Manitoba with a Bachelor of Science in mechanical engineering in 2008. Through experience and education, he has developed an expertise in building energy modelling, ground heat exchanger design, ground source heat pump systems, building commissioning, and energy auditing.



Dr. Connor Dacquay, PE, CGD

Connor is a licensed engineer in the province of Manitoba, Ontario, and New York state. He has been involved in feasibility studies and detailed design for over 30,000 tons of heat pump capacity across North America, Asia, and Australia. Connor has a Master of Science degree in civil engineering from the University of Manitoba and a doctoral engineering degree from Akita University in the Department of Earth Energy Resource Engineering.

Previous District Feasibility Studies

Berczy Glen Development, Mattamy Homes

The first phase of the Berczy Glen Development consisted of approximately 400 homes and was broken into 4 districts system modules. Each district module serviced approximately 80 to 100 homes with a single *district pipe loop* buried in the roadway right of way. Interconnection pipes facilitated transfer of energy from one district module to another. The preliminary GHX model based on providing 100% of the heating and cooling loads for each 80-home district module. The GHX consisted of 180 vertical or HDD boreholes divided into 30 GHX modules of 6 boreholes per module. 6 GHX modules were connected to each of 5 service vaults located around the district pipe loop (Figure 13). Each district module was divided into 5 district sections each with 16 homes. One service vault with GHX provided energy to and absorbs energy from homes in each section.



Figure 13: District module concept for Berczy Glen Development.

Reference: Catherine Thorn, Director, Community Energy Planning, Enwave, 1-416-338-8920, cthorn@enwave.com

MRL Triangle in Missoula, MT

With support from U.S. EPA Office of Brownfields and Land Revitalization (OBLR), ICF and GEOptimize conducted a clean energy analysis for the City of Missoula, Montana. The clean energy analysis focused on the potential for utilizing ground source heat pumps (GSHPs) in a district energy system at the Montana Rail Link (MRL) Triangle site. The impact of rooftop solar was included in the analysis. MRL Triangle is an 11.42 acre former industrial and commercial site that Missoula is considering for redevelopment.

The team evaluated four different energy scenarios:

1. Conventional dual fuel

- Space heating and domestic hot water (DHW): Natural gas
- Space cooling: Electricity

2. Conventional all-electric

- Space heating and DHW: Electrical unit heaters and baseboards and standard electric DHW tanks
- Space cooling: Electricity

3. High-performance all-electric using air source heat pump (ASHP)

- Space heating and DHW: ASHP using electricity and standard electrical DHW tanks.
- Space cooling: ASHP using electricity.

4. High-performance all-electric using GSHP

- Space heating and DHW: GSHP derived electricity
- Space cooling: GSHP derived electricity

A conceptualized mixed-use site redevelopment was modelled consisting of 14 buildings with commercial, retail, and residential space for a total of 386,538 ft² of conditioned space. Building energy modeling, ground heat exchanger (GHX) modeling, district GHX modeling, and solar PV modeling were conducted to quantify the energy consumption of each scenario and to design the solar PV and the district GSHP system. The analysis revealed that GSHPs are the most efficient way to heat and cool a building. The GSHP scenario costs more to install, but the installation costs are mitigated by lower energy demand, energy cost savings, and incentive programs. Using a District 5G energy sharing system known as an energy transfer pipe (ETP) that contains a district pipe loop with 72 vertical bores at 400 ft depth, and 3 groundwater wells with a capacity of 40 gallon per minute (gpm) could maximize the district energy system's efficiency and cost-effectiveness. The incremental capital cost of installing the GSHP system in comparison to the other scenarios was estimated at \$1,375,849. The solar modeling revealed that implementing solar energy would reduce annual energy consumption by 55% on average across all buildings.

Carbon emissions from the MRL Triangle buildings are estimated at 707 metric tons (MT) of CO₂e in the High-Performance All-Electric scenario (Scenario # 4) before GSHP and solar are added. GSHP and solar then reduce carbon emissions by a combined total of 70%, with 34% attributable to GSHP and 36% to solar.

An important note about this report is that the solar and GSHP systems are physically and economically independent. The estimated net effects of solar, which are reported separately in this report, can also be deducted from any of the scenarios to evaluate solar impacts alone, including energy use, carbon reduction impacts, and economic impacts.

Reference: Craig Schultz, Senior Advisor, Renewable Energy, ICF, 1-202-669-6200, craig.schultz@icf.com

Bemidji State University

Bemidji State University (BSU) is a public state university located in Bemidji, Minnesota, USA with over 5,400 students and 23 buildings on campus with 1,450,000 ft² of floor area. BSU embarked on a plan to achieve carbon neutrality by 2050. GEOptimize was retained to determine the feasibility of installing a district GSHP system in the facility and compared it to a business-as-usual case with replacing the current central plant natural gas heating and electric cooling system.

To quantify the individual buildings space heating and cooling loads, previously constructed building energy models using Trane Trace 700 were used. Based on the various buildings subdivided space usage, hourly energy loads from the existing energy models were scaled either up or down depending on the floor area difference and aggregated to create a new building energy profile. This method was applied to each of the 23 buildings in this analysis. Individual energy load profiles were then optimized by using energy efficiency measures of lighting upgrades and energy recovery ventilators (ERV's). These efficiency measures allowed for a reduction in overall space heating and cooling while balancing the energy loads. This resulted in a reduced capital cost for GHX and reduced energy consumption.

The energy load profiles were used to model the size and performance of the GHX using Ground Loop Design (GLD) 2016 software and thermal properties from the soil profile on campus. A horizontal directionally drilled (HDD) GHX was determined to be the most cost effective due to abundant land area on the campus and relatively low-cost drilling rates. Based on various interviews with the BSU staff and information sharing, a building priority list was developed to portion the capital cost required to begin the project. A modular construction process was applied by utilizing energy transfer pipe technology.

Based on the methodology described, in depth financial calculations were performed for each construction step based on the GHX cost, distribution piping cost, incentives, and energy cost savings. Two in depth analyses were performed, one with efficiency measures applied to the buildings and one without. The analysis without efficiency measures was deemed unfeasible due to a 40% greater GHX cost. Additionally, the impact from carbon dioxide (CO₂) emissions comparing a GSHP and conventional heating and cooling system was performed.

Reference: Roger Garton, Senior Commercial and Industrial Representative, Ottertail Power Company, 1-218-739-8542, rgarton@otpc.com.

Wallen, Australia Housing Development

This project looked at the feasibility of installing a district geothermal system at a new housing development in Wallan, Australia, which is 60 km north of Melbourne, Australia. The development consisted of 7,000 single family homes with two schools, a social centre, and a hotel. This project looked at incorporating a district geothermal system for 290 single family homes.

Due to the limited space on each individual lot for a GHX, the design consisted of one vertical GHX in each lot with a large GHX located in the parks within the development. The single vertical GHX at each lot was calculated to provide approximately 60% of the total capacity of the home. By connecting the homes to a common district geothermal system, the remaining capacity could be provided by the large GHX fields located at the parks in the development. To model the district geothermal system, TRNSYS software was used.

This study found when comparing a conventional natural gas furnace heating and electric A/C cooling system to a district geothermal system, the electrical consumption increased by 50% while reducing the electrical demand by 29%. This showed that the geothermal system has a great benefit to the electric utility by increasing their sales without having to increase their electricity production. The study also found that on average, the annual utility bills of each home would be reduced by approximately 56% when comparing to a natural gas furnace heating and electric A/C cooling system.

Reference: Max Ploumis, Engineering Manager, Direct Energy Australia, +61 3 9546 8666, max@directenergy.com.au

University of Minnesota, Morris

The University of Minnesota, Morris (UMM) is one of five universities in the University of Minnesota system. The campus currently has 39 buildings totaling 970,000 ft² of floor area situated on 100 acres of land on the eastern side of Morris, Minnesota. This analysis included 28 of the buildings on campus connected to the central heating and cooling plant. UMM has embarked on a plan to achieve carbon neutrality by 2025. GEOptimize was retained to determine the feasibility of installing a district GSHP system on the campus and compare it to the current central plant natural gas heating and electric cooling system.

To design a GSHP system, hourly heating and cooling loads of the campus were estimated based on building size and use. The loads were calibrated to the current energy consumption on campus. Individual energy load profiles were then optimized by integrating energy efficiency measures of lighting upgrades and energy recovery ventilators (ERV's). ERV's transfer energy from exhaust air to pre-heat or pre-cool incoming outdoor ventilation air to reduce the heating and cooling load in the buildings. More efficient lighting and lighting controls reduced the cooling load as less energy is added to the space. These efficiency measures allow for a reduction in overall space heating and cooling while balancing the energy loads. This resulted in a reduced capital cost for ground heat exchangers (GHX) and reduced energy consumption.

An energy transfer pipe (ETP) system moves energy from one building that is rejecting heat to the system (cooling) to another building that is extracting energy from the system (heating). It can be thought of as a conveyor belt that moves energy around the system. A ETP district style GSHP system was analysed with and without efficiency measure. The analysis without efficiency measures was deemed unfeasible due to a 33% greater GHX cost. UMM currently plans on incorporating an energy storage system for the electricity produced on campus. The effects on energy cost and CO₂ emissions from the anticipated energy storage system was analysed.

The total incremental cost for the district GSHP system including efficiency measure upgrades was \$8,446,342 with annual energy cost savings of \$310,886. The capital cost did not include the mechanical distribution systems, as this cost would be similar regardless of the heating and cooling system type. Since the construction process is split into steps, this allows for compounding energy savings as more buildings are connected to the ETP.

The current energy consumption at UMM is 1,178,516 therms and 8,188,518 kWh annually. Converting to a GSHP with efficiency measures, the campus energy demand will be 11,932,823 kWh of electricity and 27,895 therms of natural gas. The remaining natural gas used for heating buildings was not included in the analysis. UMM can currently generate on average 10,303,521 kWh annually from solar and wind. If this energy is stored, the annual energy bill at UMM will be \$93,978 which is 85% lower than what is

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currently paid annually. UMM emissions after the retrofit are projected to be 1,098 MT of CO₂ which is 87% lower than the current emissions at UMM. Incorporating the GSHP system enables UMM to use all the electricity produced on campus.

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